

2) Centerline T?

$$R_f = 0.6 \text{ cm} \quad t_f = 0.01 \text{ cm} \quad t_c = 0.06 \text{ cm}$$

$$T_{\text{cool}} = 550 \text{ K}$$

$$Q = 300 \text{ W/cm}^2 \quad K_f = 0.03 \text{ W/cm-k} \quad K_c = 0.15 \text{ W/cm-k} \quad h_{\text{cool}} = 25 \text{ W/cm}^2\text{-k}$$

outside in

$$T_{\text{co}} - T_{\text{cool}} = \frac{LHR}{2\pi R_f} \frac{1}{h_{\text{cool}}}$$

$$LHR = \pi R_f^2 Q$$

$$LHR = 339 \text{ W/cm}$$

$$\frac{LHR}{2\pi R_f} = 90 \text{ W/cm}^2$$

$$T_{\text{co}} = 90 \frac{1}{2.5} + 550 = 586 \text{ K}$$

$$T_{\text{ci}} - T_{\text{co}} = \frac{LHR}{2\pi R_f} \frac{t_c}{K_c} = 90 \frac{0.06}{0.15} = 36 \text{ K} \Rightarrow T_{\text{ci}} = 622 \text{ K}$$

$$T_s - T_{\text{ci}} = \frac{LHR}{2\pi R_f} \frac{t_f}{K_f}$$

$$K_f = 16 \times 10^{-4} T^{2.79}$$

$$K_f = 16 \times 10^{-4} (622)^{2.79}$$

$$K_f = 0.0024 \text{ W/cm-k}$$

$$T_s = 90 \frac{0.01}{0.0024} + 622 = 855 \text{ K}$$

$$T_o - T_s = \frac{LHR}{4\pi K_f} = \frac{339}{4\pi (0.03)} + 855 = \boxed{1754 \text{ K}}$$

2) empirically account for $k(T)$

$$k_0 = \frac{1}{A + BT}$$

$$A = 3.8 + 200 \text{ FIMA}$$

$$B = 0.0217$$

$$FIMA = 0$$

$$\frac{1}{\delta} \ln \left(\frac{A + BT_0}{A + BT_s} \right) = \frac{LNR}{4T}$$

$$\frac{1}{0.0217} \ln \left(\frac{3.8 + 0.0217 T_0}{3.8 + 0.0217 (855)} \right) = \frac{339}{4T}$$

$$\ln \left(\frac{3.8 + 0.0217 T_0}{22.35} \right) = 0.585$$

$$\frac{3.8 + 0.0217 T_0}{22.35} = 1.796 \quad \left(T_0 = 1674 \text{ K} \right)$$

$$\downarrow \sim 80 \text{ K}$$

3) K_{ox} @ 5% FIMA @ 1500 K

$$K = \frac{1}{A + BT}$$

$$A = 3.8 + 200 \text{ FIMA} = 3.8 + 200 (0.05) = 13.8$$

$$B = 0.0217$$

$$K = \frac{1}{13.8 + 0.0217 (1500)} = 0.0216 \text{ W/cm-k}$$

$$K_{fresh} = \frac{1}{3.8 + 0.0217 (1500)} = 0.0275 \text{ W/cm-k}$$

$\sim 22\%$ reduction

$$3) Q = U-102r$$

$$E = 1990 \quad \rho = 16 \text{ g/cc} \quad \sigma_f = 5875 \quad \phi = 5 \times 10^{14} \text{ g/cm}^2$$

$$Q = E_f \sigma_f N_f + M(U-102r) = 0.78(0.19 \times 235 + 0.81 \times 238) + 0.22(92) = 205.4 \text{ g/mol}$$

$$= 16 \text{ g/cc} \frac{1 \text{ mol}}{205.4 \text{ g}} \frac{4.022 \times 10^{23} \text{ molec.}}{1 \text{ mol}} \frac{0.794}{1 \text{ molec.}} \times 0.19 = 6.951 \times 10^{21} \text{ g/cm}^3$$

$$Q = (200 \times 10^6 \text{ eV}) (1.402 \times 10^{19} \text{ J/eV}) (587 \times 10^{24} \text{ cm}^2) (5 \times 10^{14} \text{ g/cm}^2) (6.951 \times 10^{21})$$

$$Ex = 653.7 \frac{\text{J}}{\text{cm}^2} = 653.7 \frac{\text{W}}{\text{cm}^2}$$

inverse?

given a Q, how much enrichment?

$$M(U-102r) = 0.78(x \times 235 + (1-x) \times 238) + 0.22 \times 92$$

$$N_f = 16 \text{ g/cc} \frac{1 \text{ mol}}{\text{g}} \dots$$

Forward Euler

$$\frac{dy}{dt} = -5y \quad dt = 0.33 \quad t_0 = 0 \quad y_0 = 1$$

$$y_1 = y_0 + dt y_0'$$

$$t_1 = 0.33 \quad = 1 + 0.33(-5 \times 1) = -0.65$$

$$t_2 = 0.66 \quad = -0.65 + 0.33(-5 \times -0.65) = 0.425$$

$$t_3 = 1 \quad = 0.425 + 0.33(-5 \times 0.425) = -0.275$$

1

Rad Length 3.2m $U_{H2O} = 300 \frac{W}{cm}$ $\gamma = 1.1$

L_{H2O} @ $z = 0.8$

$$Z_0 = \frac{L}{2} = 1.6m$$

$$L_{H2O}(z) = L_{H2O} \cos\left(\frac{\pi}{2\gamma} \left(\frac{z}{Z_0} - 1\right)\right)$$

$$= 300 \cos\left(\frac{\pi}{2(1.1)} \left(\frac{0.8}{1.6} - 1\right)\right) = 226.7 \frac{W}{cm}$$

coolant T @ $z = 0.8$

$$c_p = 4200 \frac{J}{kg \cdot K}$$

$$\dot{m} = 0.3 \frac{kg}{s}$$

$$T_{cool}^{in} = 500K$$

$$T_{cool}(z) = \frac{d\gamma}{\pi} \frac{Z_0 L_{H2O}}{\dot{m} c_p} \left[\sin\left(\frac{\pi}{2\gamma}\right) + \sin\left(\frac{\pi}{2\gamma} \left(\frac{z}{Z_0} - 1\right)\right) \right] + T_{cool}^{in}$$

$$= \frac{2(1.1)}{\pi} \frac{(160)(300)}{0.3(4200)} \left[\sin\left(\frac{\pi}{2(1.1)}\right) + \sin\left(\frac{\pi}{2(1.1)} \left(\frac{0.8}{1.6} - 1\right)\right) \right] + 500$$

$$T_{cool}(z=0.8) = 509K$$